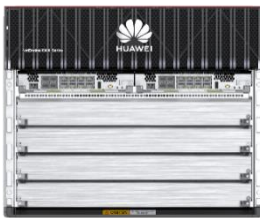


NetEngine 8000 X Series

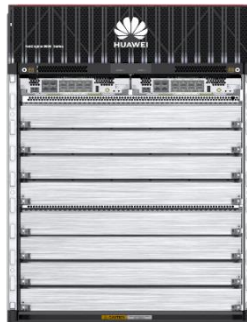
All Scenarios Intelligent Routers

Overview

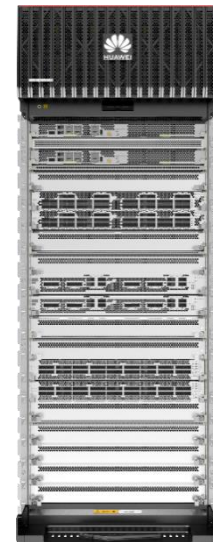
Huawei's NetEngine 8000 X series comprises the X4, X8, and X16, which are full-service intelligent metro routers designed to meet the requirements for new services and numerous connections in cloud era. The X series adopts a hardware architecture with provides a capacity of 1T per slot and it could be upgrade from 1T per slot to 19.2T per slot, and also supports SRv6 for end-to-end intelligent connections, and integrates AI-boosted intelligent O&M capabilities, helping carriers build simplified intelligent metro networks. The X series applies to metro core, aggregation, DC gateway, and international gateway scenarios.



NetEngine 8000 X4 Router



NetEngine 8000 X8 Router

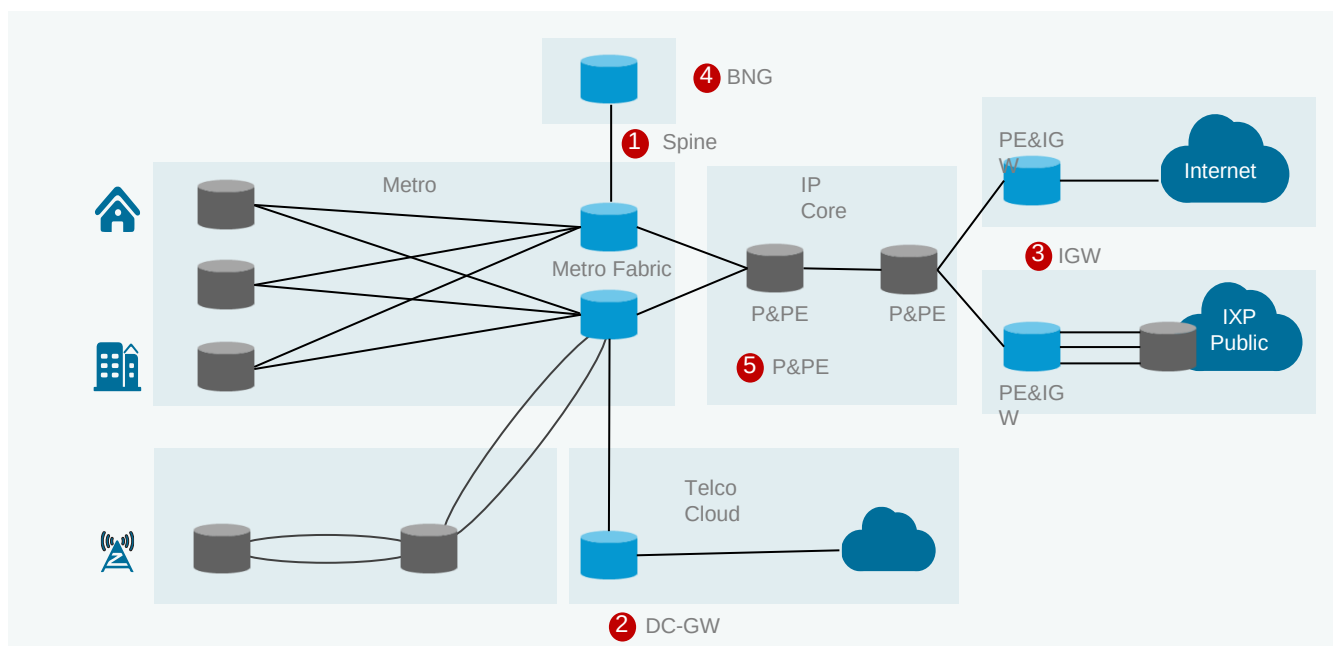


NetEngine 8000 X16 Router

Product Highlights

Highlights	Description
Super Capacity	- Large Capacity : 19.2T/slot, twice the industry average - High density board : 6*800GE, 36*400GE, 40*100GE, 72*25GE/10GE - Orthogonal architecture, saving 24% space
Various Ports	Support P/PE, SR, IGW, DC-GW and BNG 800GE/400GE/100GE/50GE/40GE/25GE/10GE/GE/FE - Support Integrated Line Processing Unit and Flexible card
All Scenarios	Metro core, Metro aggregation, DC-GW, SR scenarios, BNG, CGN, IPSEC, SA etc.

Application Scenarios



The NetEngine 8000 X series are large-capacity, highly reliable, and energy-saving routers developed by Huawei to meet the interconnection requirements of the cloud and AI era. The NetEngine 8000 X series routers can be deployed in the following positions: PE/P, Metro aggregation, Metro core, DC-GW, and IGW.

Product Specifications

Category	Specification		
	NE8000 X4	NE8000 X8	NE8000 X16
Slot Capability	1T/2T/4T/14.4T/19.2 T per slot	1T/2T/4T/9.6T/14.4T/19.2T per slot	1T/2T/9.6T/14.4T per slot
Service slots	4	8	16
Dimensions (WxDxH, mm)	442 x 872 x 437.4 (9.8U)	442 x 872 x 703 (15.8U)	442 x 1033 x 1436 (32.3U)
Chassis height [U]	9.8 U	15.8 U	32.3 U
Weight without packaging [kg(lb)]	72 kg (158.7 lb)	94 kg (207.2 lb)	244.8 kg (539.69 lb)
Weight with packaging [kg(lb)]	138 kg (304.2 lb)	177 kg (390.2 lb)	315.6 kg (695.78 lb)
Weight without packaging (full configuration) [kg(lb)]	4T bundle: 192.7 kg (424.8 lb); 14.4T bundle: 210.1 kg (463.2 lb)	4T bundle: 308.5 kg (680.1 lb); 14.4T bundle: 340.5kg (750.7 lb)	[LPUI-4T boards are fully configured]: 666.1 kg (1468.5 lb); [LPUI-14.4T boards are fully configured]: 707.7 kg (1558.9 lb)
Typical power consumption (with configuration) [W]	5240 W (estimated value when LPUI-4T boards are fully configured); 8860 W (LPUI-14.4T boards are fully configured)	10256 W (estimated value when LPUI-4T boards are fully configured)	24015 W (estimated value when LPUI-4T boards are fully configured); [LPUI-14.4T boards are fully configured]: 35965 W
Typical heat dissipation (with configuration) [BTU/hour]	17324 BTU/hour	32665 BTU/hour	[LPUI-4T boards are fully configured]: 77913 BTU/hour; [estimated value when LPUI-14.4T boards are fully configured]: 116684 BTU/hour
MTBF [year]	18.71 year	18.61 year	18.53 year
MTTR [hour]	0.5 hour	0.5 hour	0.5 hour
Availability	0.99999	0.99999	0.99999

Product Features

Category	NetEngine 8000 X4	NetEngine 8000 X8	NetEngine 8000 X16
L2	IEEE 802.1q, IEEE 802.1p, IEEE 802.3ah, IEEE 802.1ag, STP/RSTP/MSTP, G.8032, LACP, VxLAN, VLAN and bridge domain		
L3	OSPF, OSPFv3, RIP, RIPng, IS-IS, IS-ISv6, BGP, BGP4+, ACL, ACL6, 6PE, 6VPE, Static routes, GRE,		
MPLS	LDP, RSVP-TE, VPLS, VPWS, L3VPN, Seamless MPLS		
SR MPLS	ISIS/OSPF/BGP for SR MPLS, SR MPLS BE, SR MPLS TE, SR MPLS TE Policy, SR-MPLS Flex-Algo, TI-LFA, Anycast FRR, SR-MPLS Microloop Avoidance, SR-MPLS OAM		
SRv6	ISIS/OSPFv3/BGP for SRv6, SRv6 BE, SRv6 policy, SRv6 Flex-Algo, BGP Policy, SRv6 SRH Compression, SRv6 TI-LFA FRR, Anycast FRR, SRv6 Microloop Avoidance, SRv6 OAM, SRv6 SFC, APN6		
EVPN	EVPN L3VPN, EVPN VPWS, EVPN VPLS		
Multicast	IGMP, MLD, Static Multicast Routing, PIM-SM/SSM, MSDP, MBGP, Rosen MVPN, MVPN, NG MVPN, Multicast NAT, BIER, BIERv6		
Slicing	FlexE, Channelized Sub-interface, Slice-ID		
QoS	5 Level HQoS		
Clock	1588v2, Synchronous Ethernet Clock, G.8275.1, G.8275.2, SMPTE-2059		
OAM	IFIT, Telemetry, NetStream, SNMP, NQA, TWAMP, RFC2544, Y.1564, BFD, SBFD, MPLS OAM, MPLS TP OAM, VRRP, Ethernet OAM, Y.1731		
BNG	PPPoE, IPoE, L2TP , 802.1x, EDSG		
Value add services	CGN, IPSEC, SA		

Support Boards

Category	Boards	Description
Integrated Line Processing Unit	LPUI-1T3	10*100GE QSFP28 MACsec + 36*10GE(SFP+)/GE(SFP) MACsec
	LPUI-2T/ LPUI-2T-E/ LPUI-2T-L	72*25GE(SFP28)/10GE(SFP+)/GE(SFP)
		20*100GE QSFP28
		20*100GE QSFP28 MACsec
		10*100GE QSFP28+36*25GE(SFP28)/10GE(SFP+)/GE(SFP)
	LPUI-4T/ LPUI-4T-E/ LPUI-4T-L	8*400GE QSFP-DD+8*100GE QSFP28
		4*400GE QSFP-DD+24*100GE QSFP28
	LPUI-7T2	18*400GE QSFP-DD
	LPUI-14T4	36*400GE QSFP-DD
	LPUI-19T2	6 x 800GE QSFP-DD + 36 x 400GE QSFP-DD
Integrated Service Unit	/	VSUI-400-E + CGN/IPSEC/SA RTU
Flexible Card	LPUF-1T/ LPUF-1T-E	P1T: 36*10GE SFP+
		P1T: 5*100GE QSFP28
		P1T: 10*10GE + 16*155M/16*622M/8*2.5G/2*10G POS
	LPUF-2T	P1T: 36*10GE SFP+
		P1T: 5*100GE QSFP28
		P2T: 10*100GE QSFP28
		P2T: 2*400GE QSFP-DD+2*100GE QSFP28
	LPUF-6T4A	4 x 400GE QSFP-DD + 16 x 100GE QSFP28

More Information

For more information about the NetEngine 8000 X Series Routers, visit <https://e.huawei.com> or contact us in the following ways:

- Global service hotline: <http://e.huawei.com/en/service-hotline>
- Logging into the Huawei Enterprise technical support web: <https://support.huawei.com/enterprise/>
- Sending an email to the customer service mailbox: support_e@huawei.com

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